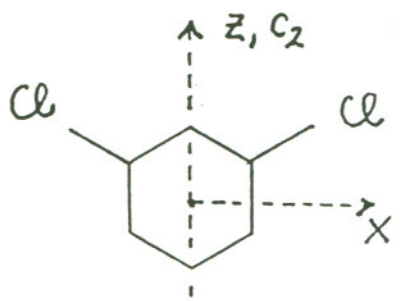
PHYSICAL CHEMISTRY CUMULATIVE EXAM

October 1, 1994

Consider point group C2v. Use a cartesian coordinate system with x,y,z axes that are oriented as follows: The z-axis is along the C2 axis, in the paper plane together with the x-axis. The y-axis is perpendicular to the paper plane. A meta-dichloro-benzene molecule, belonging to point group C2v, is shown in the Figure and placed in the paper plane. The character table is given below.

$\mathscr{C}_{2v}$	E	C_2	$\sigma_v(xz)$	$\sigma_v(yz)$
A_1	1	1	1	1
A_2	1	1	−1	−1
B_1	1	−1	1	−1
B_2	1	−1	−1	1



1) To which irreducible representations of point group C2v do the translations along the x-, y-, and z-axes belong?

2) To which irreducible representation of point group C2v does the x-component of angular momentum, J_x, belong? $J_x = yp_z - zp_y$.

3) What is the character of the two-dimensional representation based on the two C-Cl (carbon-chlorine) bond stretches of meta-dichloro-benzene? List the numbers in the sequence of E, C_2, $\sigma_v(x,z)$ and $\sigma_v(y,z)$.

4) To which irreducible representations of point group C2v do the C-Cl bond stretches of meta-dichloro-benzene belong?

5) To which irreducible representations of point group C2v do the C-H bond stretches of meta-dichloro-benzene belong?